

Evaluating the predictive power of field variables for species and individual molecular identification on wolf noninvasive samples

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Abstract Live-trapping elusive animals is often challenging, hampering the achievement of reasonable sample sizes for molecular studies. In such cases, the use of noninvasive samples (NIS) is critical in many research fields, mostly related to ecology, management and conservation of wild species. We analysed the influence of several variables potentially associated with the quality of wolf NIS—season, weather conditions, and in situ collected site and sample characteristics—on the success rates of species and individual identification performed using mtDNA and 13 microsatellites, respectively. NIS included scats, urine and saliva collected from two areas in Portugal. Scat samples exhibited the highest success rate for both species (81%) and individual identification (59%), compared with urine (63 and 30%, respectively) or saliva samples

(48 and 36%, respectively). The success rate of species identification of scats was better explained by season of collection, the presence of mucous, moisture and odour. For samples with successful species identification analysis, individual identification success was best predicted by the presence of odour. Performing a preliminary selection of scat samples with the best characteristics can increase up to 13% the success rates of molecular analysis. Urine collected on snow had a higher success rate of species identification than that collected on vegetation. To our knowledge, this was the first time that wolf urine on vegetation near ground-scratching marks is used as DNA source. Saliva samples collected with different substrate types can also be used for species identification. These results contribute to optimising noninvasive sampling procedures, maximising the success of molecular ecology studies, and ultimately minimising sampling efforts and costs.

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Introduction

The use of noninvasive genetic sampling in wildlife research and monitoring is currently of prime importance for management and conservation actions (Waits and Paetkau 2005; Beja-Pereira et al. 2009). Indeed, the use of noninvasive samples (NIS) has drastically improved, starting with species identification by amplifying and sequencing a small mitochondrial DNA (mtDNA) fragment (Palomares et al. 2002; Onorato et al. 2004; O'Reilly et al. 2008) up to the evaluation of population and ecological parameters using different molecular and statistical approaches (e.g. Lucchini et al. 2002; Marucco et al. 2009; Harris et al. 2010; De Barba et al. 2010;

Stenglein et al. 2011; Godinho et al. 2012; Czarnomska et al. 2013). Noninvasive genetic sampling provides a large amount of data without the need to capture, handle or observe animals, allowing a recurrent and cumulative use in the study of elusive carnivore species (Waits and Paetkau 2005; Kelly et al. 2012).

However, NIS are prone to misidentification among sympatric species, and their use frequently requires genetic confirmation (e.g. Davison et al. 2002; Monterroso et al. 2012). Furthermore, DNA in NIS is generally in low quantity and often contaminated and degraded due to environmental or intrinsic factors (Beja-Pereira et al. 2009; Vynne et al. 2012; Schrader et al. 2012), requiring the amplification of multiple replicas per sample to maximise success and minimise genotyping errors (Taberlet et al. 1999; Valière et al. 2002; Schwartz et al. 2006). The assessment of NIS conditions in the field could prompt an increase in the overall efficiency of the analysis. Previous studies have shown that several variables may influence the success of DNA extraction and amplification from NIS. However, these studies usually analysed few such parameters, including e.g. season, environmental factors and sample age (Piggott 2004; Wasser et al. 2004; Murphy et al. 2007; Vynne et al. 2012; Harms et al. 2015). In addition, some studies evaluated the effect of variables under controlled settings that may not accurately reflect field conditions (e.g. Santini et al. 2007). Furthermore, the results obtained regarding the influence of season on faecal DNA amplification are contradictory across studies (Lucchini et al. 2002; Scandura et al. 2006; Lonsinger et al. 2014), possibly suggesting geographic variance due to different environmental conditions. Finally, there is scarce knowledge on the influence of in situ variables such as site and sample characteristics on the success of molecular analysis for NIS.

Scats are among the most commonly used NIS for wolves because they are easy to find and provide additional biological information beyond DNA (e.g. health, physiology and diet; Barja et al. 2008; Creel et al. 2002; Darimont et al. 2008; Mech et al. 2012; Sidorovich et al. 2003; Stephenson et al. 2013). Urine deposited on snow can also be detected and collected, and is known to be a valuable source of DNA (Valière and Taberlet 2000; Hausknecht et al. 2006; Kelly et al. 2012), but its usefulness when deposited on other substrates has not been assessed. Saliva samples collected on wounds of prey have been increasingly used to investigate depredation, allowing the identification of predator species and sex (Williams et al. 2003; Blejwas et al. 2006; Sundqvist et al. 2008; Frosch et al. 2011; Caniglia et al. 2013; Mumma et al. 2014; Wengert et al. 2014; Harms et al. 2015).

In this study, we aimed to analyse the influence of a wide range of variables related to season, weather conditions and in situ collected site and sample characteristics on the success of species and individual molecular identification from wolf (*Canis lupus*) scat, urine and saliva samples.

Materials and methods

Sample collection

We collected 713 putative wolf NIS (scats, urine and saliva) between August 2006 and March 2012 in two study areas in Portugal: (1) north-western, encompassing the territory of five packs; and (2) central-northern, where six packs occur at south of the Douro river, representing 17% of the total number of packs in the country (Pimenta et al. 2005), and occupying a total area of 2358 km². Both areas have similar environmental traits, including human-dominated landscapes (69 inhabitants/km²; INE 2011), altitudes between 180 and 1370 m, and a temperate Atlantic-Mediterranean climate. Climate is characterised by a marked seasonality, with hot summers and rainy winters (1200 mm of annual average precipitation; INE 2013) with little snow (less than 30 days/year of snow cover) and average annual temperature of 14 °C (IPMA 2012).

Scats were collected during fixed monthly and seasonal transects conducted on foot or by car (<10 km/h) along unpaved roads across the study areas. Particular attention was given to usual wolf scent marking places such as paths and junctions (Barja et al. 2005; Vilà et al. 1994). Urine was collected on snow when associated with a wolf track and on vegetation when associated with scent marks (ground scratching) of a single wolf. Saliva samples were obtained from bite marks or consumption wounds from 12 predation events apparently caused by wolves in the 3 days preceding sampling or, for one sample, in the preceding week. A small quantity of substrate (i.e. snow, vegetation, hair, muscle, bone) was collected together with urine and saliva (Table 1).

Four types of variables were recorded in situ during the collection of each scat sample: (1) season, (2) weather conditions, (3) site characteristics and (4) sample characteristics (Table 1). Three categories were considered for the variable 'cover', which regards the quantity of vegetation cover directly above the scat: *covered*, when the sample was shielded from direct sunlight; *partially covered*, when only a part of the sample was exposed; and *uncovered*, when entirely exposed. Sample 'freshness' was evaluated by the combination of general appearance, presence or absence of moisture, presence or absence of mucous, and the period between consecutive samplings. Samples were considered as (1) *fresh* when intact, warm, completely moist, with fluid external mucous or detected in a place that had been sampled the previous day; (2) *middle fresh* when nearly intact, with moist only inside, with dry external mucous or less than 14 days between samplings; and (3) *old* when smashed, spread on the ground, dry, with no external mucous or at least 14 days between samplings. Scat odour was classified as present or absent to the human observer. To reduce the risk of pathogen transfer to collectors, samples were manipulated cautiously with gloves and with no

Table 1 Variable types and categories recorded for noninvasive samples (scats, urine and saliva)

Sample type	Variable type	Variables	Categories
Scat (<i>N</i> = 670)	S	Season	Spring/summer/autumn/winter
	W	Mean rainfall per day (mm) ^a	0/0.01–1.00/1.01–3.00/3.00–10.00/>10.00
	SI	Cover	Covered/partially covered/uncovered
	SA	Substrate	Snow/paved/soil/rock/vegetation
		Freshness	Fresh/middle fresh/old
		Consistency	Solid/diarrheic
		Mucous	Present/absent
		Moisture	Moist/dry
		Odour	Present/absent
Urine (<i>N</i> = 16)	SI	Substrate	Vegetation/snow
Saliva (<i>N</i> = 27)	SI	Substrate	Skin and hair/hair/muscle/bone

Variable types are referred to as S (season), W (weather conditions), SI (site characteristics) and SA (sample characteristics)

^a Mean rainfall per day information obtained from APA—Agência Portuguesa do Ambiente 2015.

direct contact with the skin when smelling. The rainfall associated to each scat was quantified by considering the total rainfall on the day of collection for *fresh* samples, the mean rainfall per day over the 8 days preceding sampling for *middle fresh* samples and the mean rainfall per day over the 30 days preceding sampling for *old* samples (Table 1).

Urine and saliva samples were collected with the substrate, and the same variables were recorded (when applicable), but among site characteristics only the type of substrate was taken into account (Table 1). All samples were collected with gloves and stored in DNA-free tubes. Scat and urine samples collected on snow were preserved in 96% ethanol, while urine samples collected on vegetation (*n* = 5) and saliva samples (*n* = 20) were preserved in silica gel.

Molecular procedures

DNA from scats was extracted following Frantz et al. (2003) after the GuSCN/silica method (Boom et al. 1990). DNA from samples preserved in silica gel was extracted following this same methodology, but with an additional prewash in PBS solution to release the cells into the solution. DNA was extracted from urine and saliva samples preserved in 96% ethanol by using the commercial QIAamp DNA Micro Kit (QIAGEN) after an overnight prewash in lyses solution and following the manufacturer's instructions. To monitor for potential DNA contaminations, negative controls were included throughout the entire process. All pre-PCR procedures were performed in dedicated laboratories used exclusively for low-quality DNA samples.

We assessed species identification through the amplification of an approximately 425 bp sized fragment of the mtDNA control region I using primers ThrL-15926 and DLH-16340

(Vilà et al. 1999). Successful amplifications were sequenced for both strands following the BigDye chemistry (Applied Biosystems) and sequencing products were separated in an ABI3130xl DNA analyser. Sequences were aligned and compared using SeqScape 2.0 (Applied Biosystems). Samples with wolf mtDNA were genotyped for a set of 13 microsatellites selected among the most variable in Iberian wolves (Godinho et al. 2011, 2015). Fluorescence labelling was achieved following Blacket et al. (2012; see Online Resource 1 for further details on markers and genotyping). Microsatellites were amplified in four multiplex reactions using the QIAGEN Multiplex PCR Kit and following the manufacturer's instructions. PCR products were separated by size on an ABI3130xl DNA analyser. Alleles were determined using GENEMAPPER 4.1 (Applied Biosystems) and checked manually.

The quality of noninvasive DNA was assessed by performing four PCR replicas for an initial set of four markers. Samples exhibiting successful amplification in this step were analysed further. After PCR amplification of all markers in a subset of 50 samples, we carried out a maximum likelihood evaluation of allelic dropout (ADO) and false alleles (FA) using PEDANT 1.0 (Johnson and Haydon 2007), and used simulations on GEMINI 1.3 (Valière 2002) to evaluate the number of genotyping replicas needed to minimise errors. This number was set to four. The consensus genotype over the four replicas for all samples was achieved following rules defined in Godinho et al. (2015). Finally, we used GIMLET 1.3.3 (Valière 2002) to compute the final error rates (ADO and FA) for each marker and across samples, to identify multiple samples from the same individual and to quantify the probability of individuals having identical genotypes (probability of identity, PID and PIDsibs).

Data analysis

Species and individual identification success rates for each sample type and variable category (Table 1) were assessed as the proportion of samples successfully identified over the total number of samples analysed for each molecular procedure. The 95% confidence intervals (CI_{95}) for each success rate were obtained according to Wilson (1927). For species identification, this rate was based on the total number of samples submitted to DNA extraction, while for individual identification it was based on the number of samples successfully screened for mtDNA. Fisher's exact tests (Fisher 1954) were performed to test whether success rate varied among the three sample types.

Samples with missing values in many (>6) variables were excluded from analysis, and the remnant missing values were replaced using K-nearest neighbours imputation method (Troyanskaya et al. 2001), performed with the VIM package (Templ et al. 2011) for R, which searches all the datasets to find the most similar samples. Due to a high quantity of missing values, 22 and 18 scat samples were excluded from GLM analysis of species and individual identification, respectively.

Success rates were transformed into logit variables and a logistic regression (Crawley 2002) with backward stepwise elimination was used to model the contribution of each predictor variable on the success of species and individual identification of scat samples. The obtained models including an intercept-only model were ranked based on Akaike's information criterion corrected for small samples (AICc) and model weights (w_i) (Burnham and Anderson 2002). The McFadden pseudo- R^2 was obtained to measure the goodness of fit of each model, and the variables included in the top 95% confidence model sets (i.e. those adding up to a cumulative weight ≤ 0.95) were considered as predictors that have a higher influence in the success of molecular analysis.

GLM analysis was performed with the MuMIn package of the R software (Bartoń 2009). We confirmed the absence of correlation between predictor variables using a chi-square test and used Cramer's V index to quantify the strength of such correlation ($V < 0.7$, $p < 0.05$; Cramer 1946; Dormann et al. 2013). Correlations among predictor variables were all lower than 0.7. Further details of correlation results are listed in Online Resource 2.

The influence of the sample preservation method on the success of species identification in urine and saliva was tested through Fisher's exact test (Fisher 1954). Success rates for species and individual identification in different substrate types on urine and saliva samples were obtained. Statistical analyses were conducted in The R Language for Statistical Programming (R Core Team 2014) and RStudio software (RStudio Team 2015), and tests were considered to be statistically significant at $p < 0.05$.

Results

We collected 713 putative wolf NIS—namely scats ($N = 670$), urine ($N = 16$) and saliva ($N = 27$). For species identification of scats, urine and saliva, we obtained success rates of 81.2% (CI_{95} 78.1–84.0%), 62.5% (CI_{95} 38.6–81.5%) and 48.1% (CI_{95} 30.7–66.0%), respectively. As expected, the success rate of species identification of scat samples was significantly higher than that observed for saliva samples (Fisher's exact test, $p < 0.001$) (Fig. 1). Scat samples were assigned to wolf (83.3%), dog (12.9%) and fox (3.9%). All urine samples and 85% of saliva samples were from wolf. For individual identification of scats, urine and saliva, we obtained success rates of 58.7% (CI_{95} 54.1–63.2%), 30.0% (CI_{95} 10.8–60.3%) and 36.4% (CI_{95} 15.2–64.6%), respectively, allowing the identification of 119 different wolves (Fig. 1). Mean values achieved for ADO and FA across loci (11.2 and 0.7%, respectively) and across samples (10.8 and 0.7%) were lower or within the values reported for canid NIS (Sundqvist et al. 2008; Bohling and Waits 2011; Caniglia et al. 2012, 2013; Godinho et al. 2015), supporting the efficiency of the selected loci. The achieved probability of identity and probability of identity between siblings ($PID = 1.27 \times 10^{-11}$ and $PIDSib = 3.22 \times 10^{-05}$) ensured that it was very unlikely to have different individuals with identical genotypes within the dataset (Waits et al. 2001).

Influence of field variables on molecular analysis: scat samples

The best GLM model for species identification of scats ($n = 648$) included the predictors of season, mucous, moisture and odour, and it exhibited a pseudo- R^2 of 0.03 (Table 2). The percentage of variation explained by these predictors was 1.5, 0.3, 0.3 and 0.6%, respectively, although only season and mucous had a significant contribution to the model (Table 3). The top 95% confidence model sets also included the predictors of freshness, consistence and cover (Table 2).

The best model for individual identification of scats ($n = 435$) showed a significant contribution of the predictor

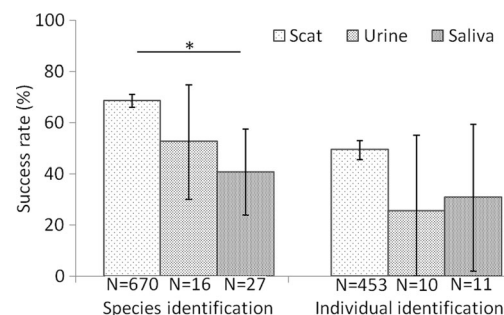


Fig. 1 Success rate (%) for species and individual identification obtained by sample type (N , total number of analysed samples). Bars refer to confidence intervals of 95% (*differences with $p < 0.05$)

Table 2 Models of scat variables for species and individual identification ranked by $\Delta AICc$

Models for Species Identification	k	LogLik	AICc	$\Delta AICc$	w_i	Cum.w	Rank
Ssn, Moist, Od, Muc	5	-308,61	631,40	0,00	0,55	0,37	1
Ssn, Moist, Fresh, Od, Muc	6	-307,25	632,80	1,39	0,27	0,70	2
Ssn, Moist, Fresh, Od, Cons, Muc	7	-307,12	634,60	3,18	0,11	0,88	3
Ssn, Moist, Fresh, Od, Cons, Muc, Cov	8	-306,20	636,90	5,48	0,04	0,96	4
Intercept-only	1	-317,78	637,60	6,17	0,03	1,00	6
Ssn, Moist, Fresh, Subst, Od, Cons, Muc, Cov	9	-303,55	640,00	8,55	0,01	1,00	7
Full model	10	-302,69	646,70	15,32	0,00	1,00	8
Models for Individual Identification	k	LogLik	AICc	$\Delta AICc$	w_i	Cum.w	Rank
Od	2	-292,71	589,45	0,00	0,38	0,38	1
Od, Cons	3	-291,86	589,77	0,32	0,32	0,69	2
Od, Cons, Muc	4	-291,78	591,66	2,21	0,12	0,82	3
Intercept-only	1	-295,02	592,05	2,60	0,10	0,92	4
Ssn, Od, Cons, Muc	5	-289,52	593,31	3,86	0,05	0,97	5
Ssn, Moist, Od, Cons, Muc	6	-289,50	595,33	5,88	0,02	0,99	6
Ssn, Moist, Fresh, Od, Cons, Muc	7	-288,85	598,22	8,77	0,01	1,00	7
Ssn, Moist, Fresh, Od, Cons, Muc, Cov	8	-288,34	601,42	11,97	0,00	1,00	8
Ssn, Moist, Fresh, Subst, Od, Cons, Muc, Cov	9	-286,64	606,57	17,12	0,00	1,00	9
Full model	10	-285,68	613,39	23,94	0,00	1,00	10

Dashed line is the limit of models included in the top 95% confidence

k number of model parameters, $LogLik$ log likelihood, $AICc$ Akaike's information criterion corrected for small samples, $\Delta AICc$ variation in $AICc$ in relation to the highest ranked model, w_i model weight, $Cum.w$ cumulative weight. Model variables: *Ssn* season, *Subst* substratum, *Fresh* freshness, *Cons* consistency, *Muc* mucous, *Cov* cover, *Moist* moisture, *Od* odour, *Rain* mean rainfall per day

of odour with an estimate of 0.464 for samples with odour ($p = 0.03$) and it exhibited a pseudo- R^2 of 0.008 (Table 2 and Table 3). The top 95% confidence model sets also included the predictors of season, mucous and consistency (Table 2).

Samples with the following characteristics had higher success rate for species identification than other categories of each variable: mucous present (83.1%, CI_{95} 78.5–86.9%), dry scats (83.2%, CI_{95} 78.9–86.8%), with odour (82.5%, CI_{95} 78.7–85.9%) and collected during summer (88.8%, CI_{95} 82.6–93.0%) (Fig. 2). Further details of the success rates of variables that had lesser influence on the GLM analysis are listed in Online Resource 3.

Influence of field variables on molecular analysis: urine and saliva samples

Preservation method did not influence the success of species identification analysis on urine and saliva samples ($p > 0.05$). We obtained a higher success rate for species identification on urine samples collected on snow than for those collected on vegetation (Fig. 3a). The opposite pattern was observed for individual identification, with a higher success rate for samples collected on vegetation than on snow (Fig. 3a).

Saliva samples collected on hair had a higher species identification success rate than on other substrates (Fig. 3b). The

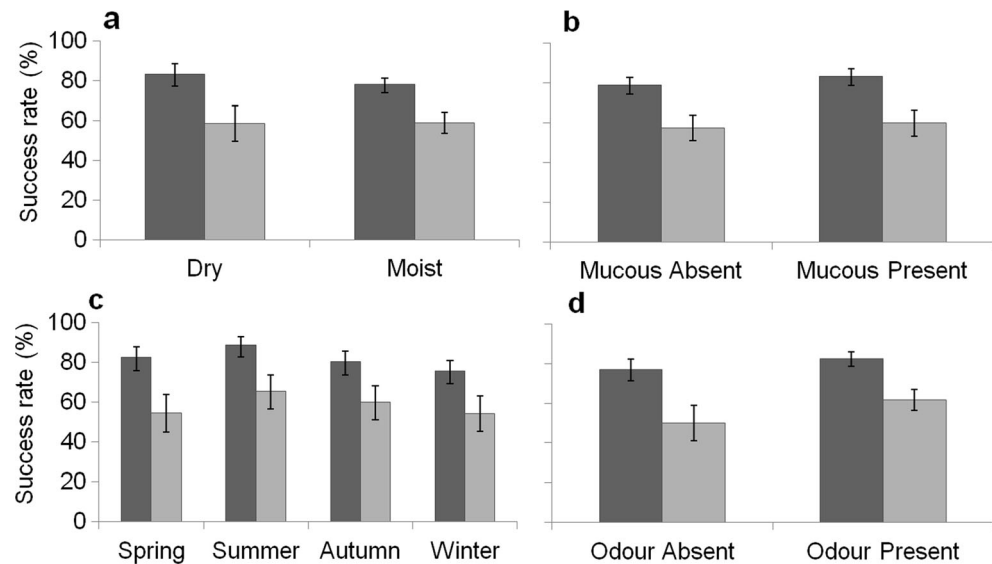
Table 3 Parameter estimates obtained by the best statistical model (lowest $AICc$), showing variables contributing to species and individual identification success on scat samples

Variable	df	SS	F	p value	η^2	Term	Estimate	SE	z value	p value
Species identification $N = 648$ —model parameters (lowest $AICc$)										
Season	3	1.52	3.315	0.020*	0.0151	(Intercept)	1.240	0.263	4.718	2.38×10^{-6} *
						Summer	0.534	0.356	1.497	0.134
						Autumn	-0.040	0.295	-0.135	0.893
						Winter	-0.357	0.269	-1.330	0.183
Odour	1	0.31	1.999	0.158	0.0031	Odour	0.364	0.218	1.669	0.952
Moisture	1	0.28	1.859	0.173	0.0028	Moist	-0.380	0.222	-1.717	0.086
Mucous	1	0.59	3.865	0.050	0.0058	Mucous present	0.438	0.215	2.036	0.042*
Individual identification $N = 435$ —model parameters (lowest $AICc$)										
Odour	1	1.13	4.684	0.031*	0.0107	(Intercept)	0.017	0.182	0.091	0.928
						Odour	0.464	0.216	2.150	0.032*

df degrees of freedom, SS sum of squares, F F value, η^2 eta-square, Term variable category, SE standard error

* $p < 0.05$

Fig. 2 Success rate for species (dark grey) and individual (light grey) identification obtained by scat samples showing the variables with higher contribution: **a** moisture, **b** mucous, **c** season and **d** odour. Bars refer to confidence intervals of 95%



success rate of individual identification was 33.3% on hair while none of the samples on muscle and on bone were suitable to be analysed (Fig. 3b). Additionally, 3 out of 14 saliva samples collected on hair exhibited contaminations from several wolf individuals (i.e. three or more alleles at a locus for several loci).

Discussion

The growing importance of NIS in wildlife research triggers the need to understand environmental and intrinsic sample features that influence the success of their use for genetic analysis. In this work, we show that selecting samples based on specific criteria related to the state of the samples during field work, as well as variables possibly related to environmental conditions, enhances species and individual molecular identifications for the NIS most commonly used in wildlife monitoring. The results of our study can serve as a basis for providing useful guidelines for both the organisation of monitoring campaigns to collect NIS, as well as for the posterior selection of the

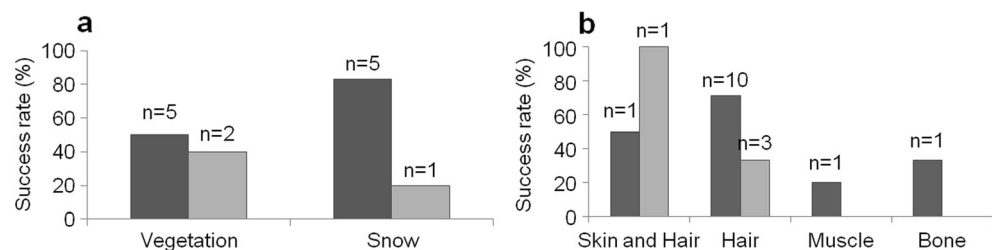
samples to be genetically analysed, thus optimising time budgets and minimising expenses.

Success determinants for scat samples

Scat samples exhibited a general success rate for species and individual identification within the values reported for other canid systems using NIS (Lucchini et al. 2002; Valière et al. 2003; Bohling and Waits 2011).

The GLM analysis indicates that few sample characteristics considered in this study had some influence on the success of species identification. The presence of mucous had a significant influence, and moisture and odour had a lower influence. Instead, we did not detect any influence of the variables related to weather conditions and site characteristics on molecular analyses. The significant positive GLM coefficient estimate obtained for samples with mucous in the success of species identification analysis was expected since it is likely that mucous has a high quantity of epithelial cells increasing the availability of DNA on the outer layer of the scat. Although non-significant, the negative GLM coefficient estimate obtained for

Fig. 3 Success rate for species (dark grey) and individual (light grey) identification obtained by **a** urine and **b** saliva samples in each substrate type (*n* number of successfully analysed samples)



moist samples in the success of species identification analysis (Table 2) may be indicative of a higher DNA quality on dry samples (Lucchini et al. 2002), and that the presence of moisture may slow or stop the process of deactivation of nucleases via the dehydration obtained through ethanol sample preservation (Beja-Pereira et al. 2009).

Interestingly, we observed that samples collected during summer performed better for species identification than those collected in winter, which contrasts with the results obtained by Lucchini et al. (2002) and Lonsinger et al. (2014). These seemingly opposite results could probably be explained by differences in winter climatic conditions between study areas. In fact, while cold and dry winters such as those reported in previous papers favour DNA preservation (Lucchini et al. 2002; Lonsinger et al. 2014), rainy and mild winters, characteristic of our study areas in Portugal, do not. Yet, a lack of association between season and DNA quality was previously observed for snowy winters in Italy (Scandura et al. 2006), supporting the need of further evaluations of this variable in regions with different climatic conditions.

In addition to season, mucous, moisture and odour, the estimates obtained for the predictors included in the top 95% confidence model sets indicate that fresher, non-diarrheic and covered samples may influence the performance on species identification analyses, though the results obtained here are inconclusive to that respect. Although sample freshness would be expected to influence species identification success rate according to previous studies on NIS of wolf (Lucchini et al. 2002; Santini et al. 2007) and other carnivores (Piggott 2004; Murphy et al. 2007; Panasci et al. 2011), we were unable to detect a significant influence of this variable. Diarrheic samples have a higher quantity of water compared with non-diarrheic ones, and less covered samples could be more exposed to environmental conditions (e.g. direct sunlight), which can accelerate the process of DNA degradation until sample collection and preservation, though no significant influence was detected in our study. We suggest that future studies should further assess the influence of these variables.

Odour was the only variable significantly influencing the success rate of individual identification. This result might be, however, biased by the quality of samples reaching this step in the analysis, which included only those which had been successfully processed for specific identification. Thus, variables significantly affecting species identification should also be considered when selecting samples for individual identification. This is the first report of odour presence positively influencing genotyping success rate in wolves, although this factor was known to be relevant for NIS quality in other carnivores like bears and foxes (Piggott 2004; Wasser et al. 2004). The estimates obtained for the predictors included in the top 95% confidence model sets (results not presented)

suggest a higher success rate on individual identification analyses of non-diarrheic samples, of samples with mucous and collected during summer. However, we could not identify a statistically significant influence of these variables, and their effect on molecular analysis should be further explored.

The low percentages of total variation explained by the best GLM models (3.0 and 0.8%) suggest that among the variables investigated here, those selected have a weak effect and that other important factors affecting the success of species and individual identification of scats were not considered. Furthermore, the low variability explained can also be related to other variables extrinsic to the sample, as uncontrolled changes in laboratory procedures, low quantity of alcohol for scat preservation or even the time between sample collection in the field and their molecular analysis in the laboratory. Thus, we suggest that future studies should consider other weather conditions, site and sample characteristics, such as scat composition, diet, time between sample collection and molecular analysis, or sample conservation method.

We suggest that in study areas with rainy and mild winters, species identification analysis should rather use scat samples collected during summer, with mucous, and preferably, with odour and being dry; when the aim is species and further individual identification, scat samples should have odour. The selection of samples that fulfil these criteria can increase the success of species and individual identification up to 13%. Indeed, when scat samples with the best variables—with mucous and odour, dry and collected during summer—are selected to be submitted to species identification analysis, the success rate increased from 81.2% (success rate without a preliminary selection, i.e. when all samples are analysed) to 94.4%. When only samples with odour are selected among those with successful species identification, the success rate of individual identification increased from 58.7 to 67.1%.

Success determinants for urine and saliva samples

Urine samples had a lower amplification success rate than scat samples, as it has been described before (e.g. Hedmark et al. 2004). To the best of our knowledge, we present for the first time and evaluate an alternative source of urine samples from wolves, consisting of urine collected on vegetation near ground-scratching marks. This innovative and valuable source of DNA is especially relevant in areas with limited snow cover, such as in the southern areas of wolf range (Blanco and Cortés 2012; Llaneza et al. 2014), or eventually in northern regions where snowfall is progressively becoming scarcer due to climate change effects (IPCC 2007; EEA 2008). Although scratching marks are not frequently observed in our study area, the frequency of urination is known to be higher than that of defecation (Zub et al. 2003; Hausknecht et al. 2006). Experienced observers or use of detection dogs to find urine on vegetation should provide an opportunity to increase the

sample size independently of snow occurrence. Our results suggest that snow seems to be preferable for species identification compared to vegetation, though the low number of samples analysed precludes us from extracting definitive conclusions. Some reasons for this result may include (1) a faster degradation of urine on vegetation, (2) lower amount of sample collected on vegetation because leaves are only sprayed with urine and (3) possible presence of PCR inhibitors in plant material, such as polyphenols (Schrader et al. 2012).

Saliva is described as a source of high-quality DNA (Abraham et al. 2012; Lobo et al. 2015), but when used in forensic scenarios, such as in our study, it raises the same concerns, or even additional ones, as scats and urine. In forensic cases, saliva is generally present in very low quantities, mixed up with different prey tissues and degraded due to environmental exposure. Indeed, saliva samples exhibited the lowest success rate for species identification, lower than those reported in most studies (55%, Caniglia et al. 2013; 54%, Mumma et al. 2014; 39%, Wengert et al. 2013). However, our results show that it is possible to perform successful species identification of saliva collected on several types of substrates, including muscle, bone, skin or hairs of prey. Further studies should be performed to evaluate the success of species and individual identification on urine and saliva samples on different substrates.

An issue associated with wildlife urine and saliva sampling is cross-contamination by several individuals, as observed in our work. This is difficult to avoid in a social species like the wolf, in which several pack members can mark with urine in the same place (Peters and Mech 1975; Asa et al. 1985; Asa et al. 1990) and participate in the attack and consumption of a single prey. Collecting several separated samples from various leaves, when sampling urine on vegetation, and from different bite marks and consumption wounds, when sampling saliva, may increase the probability of obtaining non-contaminated DNA from several individuals. Furthermore, collecting saliva samples directly near evident bite wounds could reduce the possibility of cross-contamination. However, further studies with larger sample sizes should be conducted to better evaluate the usefulness and reliability of these procedures.

Improving study design by selecting the best samples

Our results provide guidelines and suggestions for studies using scat, urine and saliva samples as noninvasive wolf DNA sources. Future molecular studies across the wide wolf range with climatic conditions similar to our study areas will benefit from taking into consideration the set of variables identified as significant determinants of genotyping success. Selecting samples with the mentioned characteristics will avail to achieve higher success rates in species and individual identification. Although shedding light on new procedures for collecting NIS, the information obtained from the analysis of

urine and saliva samples needs to be supported by additional data.

Our results contribute to the optimisation of different kinds of noninvasive sampling and prompt their use in wildlife monitoring and molecular ecology studies, especially in environmental conditions similar to Southern Europe. We argue that it is possible to optimise costs by reducing the sampling effort—decreasing the number of collected samples, effort, fuel and labour costs spent in prospecting—as well as the laboratorial analysis—decreasing the number of unsuccessful and repeated analysis and consequently the time, material and labour costs.

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Compliance with ethical standards

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Conflict of interest The authors declare that they have no conflict of interest.

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